

Bubbles POC — Manual Testing Checklist

Use this checklist to verify the full POC build before declaring Phase 6 complete.
Run through every section. Check each box only after observing the expected behavior.
If a check fails, file a bug with the section ID and expected vs actual behavior.

Environment

#	Check	Command / Step
E1	Node $\geq$ 22 installed	<code>node -v</code>
E2	pnpm 10 installed	<code>pnpm -v</code>
E3	Dependencies installed	<code>pnpm install</code> completes without errors
E4	Native binaries healthy	<code>pnpm bubbles:doctor</code> reports all green
E5	TypeScript compiles	<code>pnpm typecheck</code> exits 0
E6	Lint clean	<code>pnpm lint</code> exits 0 (warnings OK)
E7	Unit tests pass	<code>pnpm test</code> — all 10 suites green
E8	Coverage $\geq$ 70%	<code>pnpm test:coverage</code> — no threshold failures

Section A: App Launch & Smoke

Prerequisites: `pnpm --filter @bubbles/desktop build` completed successfully.

#	Check	Expected Result
A1	Launch app	<code>pnpm dev:desktop</code> — app opens within 5s
A2	Avatar window visible	256×280 transparent window appears on desktop
A3	Avatar animates	Bubbles idles with blinking animation within 3s

#	Check	Expected Result
A4	No DevTools in packaged build	<code>pnpm package:win</code> → <code>run .exe</code> → no DevTools window
A5	System tray icon	Tray icon visible in system tray area
A6	Tray menu works	Right-click tray → Show/Hide, Open Chat, Shortcuts, Quit all functional

Section B: First-Run Onboarding

Prerequisites: Fresh user (delete `~/.bubbles/` or use temp profile).

#	Check	Expected Result
B1	Wizard appears on first launch	Full-screen onboarding overlay visible with "Welcome to Bubbles"
B2	Step 1 → Next	Click "Get Started" → advances to API key step
B3	Step 2 validates key	Enter valid MiniMax key → "Checking..." → advances to folder step
B4	Step 2 rejects bad key	Enter invalid key → error message shown, stays on step 2
B5	Step 3 pick folder	Click "Browse" → folder picker dialog opens
B6	Step 3 finish	Click "Finish" → wizard closes, chat panel usable
B7	Onboarding skipped on relaunch	Quit + relaunch → wizard does NOT appear
B8	Test mode bypass	<code>BUBBLES_TEST_MODE=1 pnpm dev:desktop</code> → wizard completes with any key

Section C: Chat & Agent Interaction

Prerequisites: Onboarding complete, valid API key set (or `BUBBLES_TEST_MODE=1`).

#	Check	Expected Result
C1	Open chat via click	Click avatar → chat panel opens anchored to avatar
C2	Open chat via shortcut	Press <code>Ctrl+Space</code> → chat panel toggles
C3	Close chat via Escape	Press <code>Escape</code> in chat → chat panel hides
C4	Send text message	Type "hi" → Enter → message appears in chat
C5	Receive streaming response	Assistant reply streams in word-by-word
C6	Response within 2s	First token visible within 2 seconds of send
C7	TTS plays	Audio plays after text completes (lip-sync visible)
C8	Lip-sync active	Avatar mouth moves in sync with audio amplitude
C9	Mood transitions	<code>thinking</code> → <code>talking</code> → <code>idle</code> observed during turn
C10	Tool-call chips	When agent uses tool, small pill appears (e.g.  <code>plan_mode</code>)
C11	Tool-result chips	Tool result updates pill to  <code>plan_mode</code> or  <code>plan_mode</code>
C12	Error toast	On error, toast notification appears (auto-dismisses 4s)
C13	Cost cap blocks	Set cap to \$0.01 → send message → "I've hit today's spend cap"
C14	Cost cap allows	Reset cap to \$5 → send "hi" → normal response

Section D: Agent Switching

#	Check	Expected Result
D1	Open switcher	Click agent name in chat header → dropdown opens
D2	All 3 agents visible	Bubbles, Coda, Sage all shown with role labels and tint dots
D3	Switch to Coda	Click Coda → active agent changes, avatar tint switches to blue
D4	Switch to Sage	Click Sage → active agent changes, avatar tint switches to green

#	Check	Expected Result
D5	Switch back to Bubbles	Click Bubbles → active agent changes, avatar tint switches to default
D6	Skin persists	Quit + relaunch → last selected agent restored
D7	Coda personality	Ask Coda to write code → response is coding-focused
D8	Sage personality	Ask Sage a research question → response is thoughtful/analytical
D9	Bubbles personality	Ask Bubbles a general question → response is friendly/concise

Section E: Tools & Permissions

#	Check	Expected Result
E1	Coda plan_mode	Ask "make todo.md with 3 tasks" → <code>plan_mode</code> permission modal appears
E2	Plan rendered nicely	Modal shows steps as numbered list (not raw JSON)
E3	Approve plan	Click "Approve" → <code>writeFile</code> permission modal appears
E4	WriteFile diff preview	Modal shows file content preview (new file = "Create new file")
E5	Approve write	Click "Approve" → file created on disk at workspace root
E6	Verify file	<code>cat ~/.bubbles/workspace/todo.md</code> (or chosen root) → file exists with content
E7	Deny write	Repeat → click "Deny" → red toast "File write denied", no file created
E8	Auto-allow readFile	Ask "read README.md " → no permission modal, file read directly
E9	Remember choice	Approve + check "Remember" → same tool next time skips modal

#	Check	Expected Result
E10	Permission audit	<code>permissions</code> table in SQLite has rows for all decisions

Section F: Conversation History

#	Check	Expected Result
F1	Messages persist	Send messages → quit app → relaunch → messages visible
F2	Last conversation loaded	Chat panel shows prior conversation on relaunch
F3	Tool calls in history	Prior tool calls shown as grey <code>[tool: ...]</code> chips
F4	Multi-turn context	Ask follow-up question → agent remembers prior context
F5	Conversation per agent	Switch agent → new conversation starts
F6	Agent conversation isolated	Bubbles conversation not mixed with Coda conversation

Section G: Settings & Spend Dashboard

#	Check	Expected Result
G1	Open settings	Click gear icon in chat header → Spend Dashboard modal opens
G2	Today spend visible	Card shows today's spend (should be > 0 if messages sent)
G3	7-day history visible	Card shows last 7 days aggregate
G4	30-day history visible	Card shows last 30 days aggregate
G5	Progress bar	Today's spend vs cap shown as progress bar
G6	Change cost cap	Edit cap input → save → new cap enforced on next turn
G7	Cap persists	Quit + relaunch → cap value remembered

Section H: Avatar Behavior

#	Check	Expected Result
H1	Drag avatar	Click and drag avatar → window moves with cursor
H2	Click-through transparent	Click transparent area of avatar → click passes to window behind
H3	Avatar follows on move	Drag avatar while chat open → chat panel repositions to follow
H4	Idle animation	Avatar blinks periodically while idle
H5	Thinking animation	While agent processes → thinking animation plays
H6	Talking animation	While TTS plays → talking animation + lip-sync
H7	Error mood	On agent error → avatar briefly shows error mood
H8	Multi-monitor	Move avatar to second monitor → position persists on relaunch

Section I: Deterministic Test Mode

#	Check	Expected Result
I1	Test mode chat	<code>BUBBLES_TEST_MODE=1 pnpm dev:desktop</code> → send "hi" → canned response
I2	Test mode TTS	TTS plays silent MP3 (no audio but lip-sync fires)
I3	No API key needed	Test mode works without valid MiniMax key
I4	E2E smoke passes	<code>BUBBLES_TEST_MODE=1 pnpm e2e</code> → 5/5 specs pass
I5	Eval harness runs	<code>pnpm eval:ci</code> → 90 prompts scored, report printed

Section J: Packaging & Distribution

#	Check	Expected Result
J1	Build succeeds	<code>pnpm --filter @bubbles/desktop build</code> → no errors
J2	Package Windows	<code>pnpm package:win</code> → <code>.exe</code> produced in <code>apps/desktop/release/</code>
J3	Installer < 200 MB	<code>.exe</code> file size < 200 MB
J4	Installer runs	Double-click <code>.exe</code> → installer opens, completes
J5	Installed app launches	Start menu / desktop shortcut → app opens
J6	First-run on install	Fresh install → onboarding wizard appears
J7	No DevTools in package	Packaged app → no DevTools window
J8	Assets included	Sprites, tray icon, sounds all load correctly

Section K: CI / Automation

#	Check	Expected Result
K1	Verify workflow	<code>pnpm verify:quick</code> → typecheck + lint + test all green
K2	E2E workflow	<code>pnpm e2e</code> → builds desktop + Playwright passes
K3	Eval workflow	<code>pnpm eval:ci</code> → deterministic eval completes
K4	Coverage threshold	<code>pnpm test:coverage</code> → all packages ≥ 70%
K5	GitHub Actions verify	Push to PR → verify workflow green
K6	GitHub Actions e2e	Push to PR → e2e workflow green (Windows runner)

Section L: Edge Cases & Recovery

#	Check	Expected Result
L1	No API key	Launch without key → onboarding forces step 2
L2	Invalid API key	Enter bad key → validation fails, error shown
L3	Network offline	Live mode + no network → error toast, text still shown
L4	TTS failure	TTS error → text reply delivered, toast "Speech synthesis failed"
L5	Abort mid-turn	Close chat during streaming → turn ends cleanly
L6	Rapid messages	Send multiple messages quickly → queued, processed sequentially
L7	Long message	Paste 1000+ chars → sends, truncates history if needed
L8	Special characters	Send emoji, unicode, code blocks → renders correctly
L9	Empty message	Press Enter with empty input → nothing sent
L10	Very long conversation	50+ messages → app remains responsive

Sign-off

Role	Name	Date	Result
Tester			☐ Pass / ☐ Fail
Notes			

Total Checks: 96

Required for Phase 6 Exit Gate: All Section A–F checks + at least 80% of all checks

Blockers: Any crash, data loss, or security issue is an automatic blocker regardless of score.